Project Report

2023ICTS143

STUDENT MANAGEMENT SYSTEM

C++ Programming Project Report

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Programming Language: C++

1. Introduction

The Student Management System is a console-based application developed using C++. It helps users manage basic student information through a simple menu-driven interface.

The system allows users to add new students, display student details, search for a student using a student ID, and delete student records.

2. Objectives

The main objectives of this project are:

- To develop a simple student management application.
- To practice C++ programming.
- To understand functions, structures, arrays, loops, and conditional statements.
- To perform basic operations on student records.

3. Technologies Used

- Programming Language: C++

- Code Editor: Visual Studio Code

- Compiler: GCC (g++)

4. System Features

Add Student: Allows the user to enter a student's ID, name, course, and marks.

Display Students: Displays the details of all students currently stored in the program.

Search Student: Searches for a student using their student ID.

Delete Student: Removes a student's record using their student ID.

Exit: Allows the user to close the program.

5. Programming Concepts Used

- Structures ("struct") to group student information.

- Arrays to store student records.

- Functions to separate program tasks.

- Loops to repeat operations and search records.

- Conditional statements to make decisions.

- A switch statement to process menu choices.

6. How the System Works

When the program starts, it displays a menu with five options. The user selects an option by entering a number.

The program performs the selected operation and displays a message or the relevant student information. The menu continues to appear until the user selects the Exit option.

7. Testing and Output

The following operations can be tested:

1. Add a student with valid details.
2. Display the student information.
3. Search using an existing student ID.
4. Search using a non-existing student ID.
5. Delete an existing student.
6. Try to add a duplicate student ID.
7. Exit the program.

Output Screenshots: Insert screenshots of the program menu, adding a student, displaying student details, searching, and deleting a student.

8. Limitations

- The program stores up to 100 student records.
- Student information is stored temporarily in memory.
- Records are not saved permanently after the program closes.
- The application uses a console interface.

9. Future Improvements

In the future, the system could be improved by adding permanent data storage, a graphical user interface, student record updating, and input validation.

10. Conclusion

This project helped me practice C++ programming and understand how to organise data and perform basic operations using functions, structures, and arrays. It also gave me practical experience in developing and testing a simple software application.